rate, memory lifetime, and gate/measurement fidelity. Unlike classical repeaters, which simply amplify signals, quantum repeaters must preserve the delicate quantum states involved, often through processes like entanglement swapping and purification. The deployment of quantum repeaters not only enables longer quantum links but also enhances network security. Quantum communication is inherently resistant to interception, as any attempt to measure or tamper with a quantum signal disrupts its state and is immediately detectable. This property makes quantum networks exceptionally secure against traditional cyber threats.

Communication protocols: Sending qubits over distance

Transmitting quantum information, or qubits, across a network requires innovative communication protocols that address the unique challenges of quantum mechanics. Unlike classical bits, qubits cannot be copied due to the no-cloning theorem, necessitating the use of quantum teleportation and entanglement-based schemes. Quantum teleportation involves transferring the state of a qubit from one location to another via shared entanglement and classical communication. Protocols such as quantum key distribution (QKD) harness these principles to enable secure exchange of cryptographic keys. However, transmitting qubits over optical fibres or free-space links is subject to loss and decoherence, driving the development of robust error correction techniques and adaptive routing strategies to ensure reliable communication across quantum networks.

Early experiments and current projects

Pioneering experiments in quantum networking have demonstrated the feasibility of entanglement distribution and quantum communication over metropolitan and intercontinental distances. Notable achievements include the creation of quantum links between laboratories, the successful transmission of entangled photons via satellite, and the implementation of quantum key distribution in real-world settings. Current projects, such as the European Quantum Internet Alliance and the Quantum Internet Initiative in India, are working towards the establishment of prototype quantum networks and the development of scalable infrastructure. These initiatives bring together academic institutions, industry partners, and government agencies to advance the practical deployment of quantum communication technologies and lay the foundation for a future quantum internet.

Applications for secure and fast communication

The quantum internet holds immense promise for secure and rapid communication across a variety of domains. Quantum key distribution enables the exchange of cryptographic keys with absolute security, making it invaluable for banking, defence, and diplomatic communications. The interconnection of quantum computers via quantum networks could facilitate distributed quantum computing, allowing complex computations to be performed collaboratively across multiple nodes. Furthermore, quantum sensing and metrology applications benefit from the precision and sensitivity afforded by quantum correlations, enabling breakthroughs in scientific measurement and environmental monitoring. As quantum networking matures, it is poised to underpin a new generation of secure, high-speed communication systems that transcend the limitations of classical technologies.

Hurdles to building a scalable quantum internet

Realising a scalable quantum internet entails concurrently addressing challenges in physics, device engineering, and network architecture. The first hurdle is loss and decoherence: photons are absorbed in fibre (particularly around 1550 nm), while matter qubits (NV centres, trapped ions, neutral atoms, superconducting qubits via transduction) have limited coherence times and control errors. Unlike the classical counterparts, quantum states cannot be amplified or copied (no-cloning), thus the scaling relies on entanglement distribution with high-fidelity and verifiable quality-of-service (QoS). And that immediately gives us the second obstacle: quantum repeaters are not a single device but an integrated stack with one small cube on top of the other — memory, photonic interface, entanglement swapping, purification or error-corrected links and synchronisation — each with demanding constraints; none of them has any margin at all with respect to efficiency, dark counts, indistinguishabilty, or jitters in time.

A third challenge is heterogeneity: the networks of the future will be a hybrid of fibre, free-space and satellite links, connecting heterogeneous nodes. This calls for interoperable quantum link layers (entanglement generation protocols heralding multiplexing) as well as control-plane co-design to integrate classical signalling with quantum operations (feedforward, scheduling, buffering under memory constraints). Fourth, there is a substantial systems-engineering gap in benchmarking: a network must reveal metrics such as entanglement generation rate (EGR), fidelity, latency, and secret-key rate, in the presence of noise and adversarial behaviour. In addition, strong security requires more than 'physics guarantees': implementations need to consider side channels, device imperfections and denial-of-service threats, leading to techniques such as measurement-device-independent QKD, device characterisation and hardware-rooted trust models for quantum nodes.